

Data Professional Survey Breakdown

Project Report | Power BI

1. Project Overview

This project analyzes survey responses from data professionals to explore job roles, salary patterns, programming language preferences, geographic distribution, and career satisfaction. The dashboard summarizes the survey in Power BI and helps users compare the reported experiences of people working in, or seeking to enter, the data field.

2. Dataset Summary

- Survey responses shown in the dashboard: 630
- Key fields represented: job title, salary, country, age, favorite programming language, work-life balance satisfaction, salary satisfaction, and perceived difficulty of entering the data field.
- The dashboard includes comparisons across job titles and programming language preferences, along with summary indicators and respondent sentiment.

Note: The dashboard export does not display a complete dataset schema or detailed data-quality statistics, so the report does not assume a specific number of source columns or missing values.

3. Data Preparation and Analysis Approach

The project uses Power BI to organize and present survey data through summary cards, bar charts, a treemap, a doughnut chart, and gauge visuals. The report focuses on the fields and results visible in the supplied dashboard. The source material provided for this report does not show separate Python or SQL analysis, so those steps are not included as completed project work.

- Compared average salary across the job-title categories displayed in the dashboard.
- Compared counts of respondents by favorite programming language, with job title used as a breakdown.
- Reviewed the geographic distribution of survey respondents.
- Displayed average age and the number of survey takers as high-level indicators.
- Summarized work-life balance and salary satisfaction using 0–10 gauge visuals.
- Examined responses about the perceived difficulty of entering the data field.

4. Key Findings from the Dashboard

4.1 Survey profile

The dashboard reports 630 survey takers and an average respondent age of 29.87. The country treemap shows the United States as a prominent respondent group, with India and other country categories also represented.

4.2 Salary by job title

In the salary comparison, Data Scientist is the highest category shown, followed by Data Engineer and Data Architect. Data Analyst, Database Developer, and Student/Looking for Work appear lower in the displayed ordering. The dashboard does not provide clearly readable exact salary values for every role, so this report describes the relative ordering rather than assigning unverified figures.

4.3 Programming language preferences

Python has the largest bar by respondent count in the programming-language chart. R and the Other category follow, while C/C++, JavaScript, and Java have smaller counts in the displayed chart. This indicates that Python was the most frequently selected language among the options represented in this survey.

4.4 Satisfaction indicators

The average work-life balance rating is 5.74 out of 10, while the average salary satisfaction rating is 4.27 out of 10. Within this dashboard, salary satisfaction is lower than work-life balance satisfaction.

4.5 Perceived difficulty of entering data

The doughnut chart shows 269 respondents (42.7%) selected “Neither easy nor difficult,” 156 (24.76%) selected “Difficult,” 134 (21.27%) selected “Easy,” and 44 (6.98%) selected “Very Difficult.” The remaining category is “Very Easy”; its count is visible as a small segment but is not clearly labelled in the supplied export.

5. Dashboard in Power BI

The dashboard brings the survey indicators and charts together on one report page. It includes a country treemap, average salary by job title, survey-taker count, average age, favorite programming language, difficulty entering data, work-life balance rating, and salary satisfaction rating.

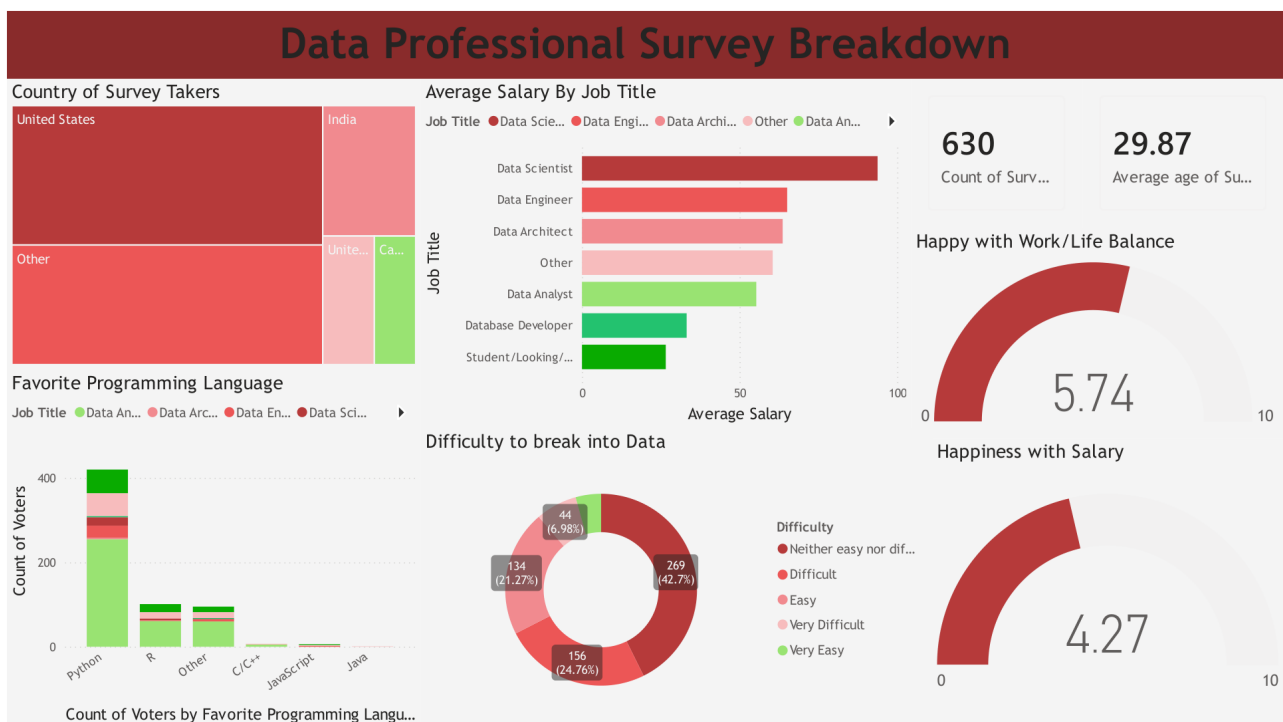


Figure 1. Dashboard supplied for the Data Professional Survey Breakdown project.

6. Business Recommendations

Career research: Aspiring data professionals can use the role comparison as a starting point for exploring career paths. Salary values should be checked against current market data before making career decisions.

Skills planning: Since Python is the most frequently selected language in this survey, learners may consider including Python in their study plan alongside role-specific tools and skills.

Salary satisfaction: The lower salary satisfaction score suggests that compensation is an area worth examining further. A follow-up analysis could compare satisfaction by role, country, and experience level.

Work-life balance: The work-life balance score is higher than the salary satisfaction score, but it is still a mid-range rating. More detailed analysis by role and industry could help explain differences.

Career entry support: A large share of respondents described entering the data field as neither easy nor difficult, while others reported difficulty. Training providers could investigate common barriers and offer practical projects, mentoring, and job-readiness support.

Data limitations: These findings describe survey respondents only. They should not be treated as a representative picture of every data professional, and the dashboard does not establish that one factor causes another.

7. Conclusion

The Power BI dashboard provides a compact view of survey responses on roles, salaries, programming language preferences, satisfaction, and perceived career-entry difficulty. Its clearest visible patterns are the leading position of Python in language preferences, the higher displayed salary category for Data Scientists, and the difference between average work-life balance and salary satisfaction ratings. Further analysis using the underlying dataset could validate these patterns and explore them by additional segments.

Project Tools

Power BI Desktop; dashboard visualizations and summary indicators.